

Tuesday Warm-up: Electrostatics

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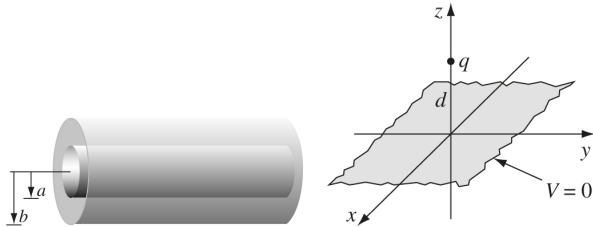


Figure 1: (Left) A cylindrical coaxial cable with capacitance per unit length. (Right) A charge above a grounded conducting plane.

1 Memory Bank

- Definition of potential

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}' \quad (1)$$

- Gauss' Law, integral form

$$\oint \mathbf{E} \cdot d\mathbf{a} = Q_{\text{enc}}/\epsilon_0 \quad (2)$$

- Electrostatic energy, continuous charge distribution:

$$W = \frac{1}{2} \epsilon_0 \int E^2 d\tau \quad (3)$$

- Capacitance, C :

$$C = Q/V \quad (4)$$

2 Capacitance

1. Consider the conductors in Fig. 1 (left). Show that the capacitance per unit length is $C/l = 2\pi\epsilon_0/\ln(b/a)$.
2. Recall that the capacitance of two identical parallel conducting plates with area A and separated by a distance d is $C = \epsilon_0 A/d$. Show that the stored energy at a potential difference V is $W = (1/2)CV^2$.

3 Potentials

1. **Uniqueness theorems.** *Uniqueness theorems* for Laplace's equation ($\nabla^2 V = 0$) and Poisson's equation ($\nabla^2 V = -\rho/\epsilon_0$) demonstrate that if a solution for the potential in a region is found, subject to boundary conditions and source charge distributions, it is the *only* solution. (a) How many solutions are there for the following differential equation?

$$\frac{d^2 V}{dx^2} = 0 \quad (5)$$

(b) Suppose $V(0) = 1$ Volt, and $dV/dx = 2$ Volts per meter. What is $V(3)$? Is the formula you used to evaluate this the only solution?

2. Consider Fig. 1 (right). A point charge q is located a distance z above the xy -plane, which is filled with a grounded conductor. (a) Write equations that describe the boundary conditions. (b) Suppose, instead of a grounded conducting plane at $z = 0$, there was a charge $-q$ a distance $-z$ below the xy -plane. Does this configuration satisfy the boundary conditions of part (a)? (c) What is the potential $V(x, y, z)$ for the two charges? Given the uniqueness theorem, this must be the *only* solution. Show that configuration of two point charges has the right units and limits.

3. Recall that $\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \sigma/\epsilon_0 \hat{\mathbf{n}}$, which implies $\sigma = -\epsilon_0(\partial V/\partial n)$ for Fig. 1 (right). Calculate $\sigma(x, y)$ and the total charge.